



Royal University
of Phnom Penh

ITE INFORMATION
TECHNOLOGY
ENGINEERING

AI SCAM SHIELD

Advisor: Mr. Chhim Bunchhun

Presented by: Yoeurn Yan





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INTRODUCTION

In the digital age, spam and phishing attacks have become increasingly sophisticated, posing significant threats to individuals and organizations. Cambodia has seen a rise in SMS scams, fraudulent messages, and phishing attempts targeting mobile users. Traditional rule-based spam filters are no longer sufficient to combat these evolving threats, creating a need for intelligent, AI-powered detection systems.





PROBLEM STATEMENT

- Increasing volume of spam messages and phishing attacks targeting mobile users
- Traditional spam filters fail to detect sophisticated social engineering attacks
- Lack of accessible, user-friendly tools for Cambodian users to verify suspicious messages
- Voice-based scams are emerging but few solutions address voice message analysis
- Users often fall victim to brand impersonation and URL-based phishing attacks



AIM AND OBJECTIVE

AIM

develop an AI-powered mobile application that detects spam, phishing, and social engineering threats in text and voice messages with high accuracy.

OBJECTIVE

- Provide real-time spam detection with 95%+ accuracy
- Detect phishing attempts including URL analysis and brand impersonation
- Support both text and voice message scanning
- Offer user-friendly mobile interface for easy threat verification
- Maintain scan history for user awareness and learning



LIMITATION AND SCOPE

LIMITATIONS:

- Platform: Mobile application (iOS and Android via Flutter)
- Target: Individual users (not enterprise-level deployment)
- Language: Primarily English text analysis



LIMITATION AND SCOPE

SCOPES (FEATURES):

Core scam detection (ML-powered):

- Text message spam detection (TF-IDF + Logistic Regression)
- Voice message transcription & analysis (multi-modal: speech-to-text + audio embeddings + prosody)
- Phishing URL & content detection, including brand impersonation & homoglyph/lookalike detection
- Email auto-scan over IMAP (Gmail / Outlook / Yahoo / custom), with scheduled background scanning

Security operations (SOC dashboard):

- Threat, incident & alert management
- Automated response playbooks
- Network activity monitoring
- User behavior anomaly analysis
- File scanning (scan-by-ID & statistics; file upload + VirusTotal lookup planned)

Platform & data:

- Scan history management
- User threat reporting system
- Statistics & analytics dashboard
- Continuous learning: user feedback → model retraining loop



LITERATURE REVIEW

Author	Method	Description	Result
Almeida et al. (2011) SMS Spam Collection	Naive Bayes,SVM,	Created UCI SMS Spam dataset with 5,574 messages	97.5% accuracy with SVM
Sahami et al. (1998) - Bayesian Spam Filtering	Naive Bayes	Probabilistic approach to email spam classification	Foundation for modern spam filters
Mohammad et al. (2014) - Phishing Detection	Machine Learning + URL Features	Feature extraction from URLs for phishing detection	92% detection rate
Gupta et al. (2021) - Deep Learning Spam	LSTM, BERT	Transformer-based text classification for spam	98.2% accuracy on benchmark
Vaswani et al. (2017) - Transformers	Attention Mechanism	Self-attention for NLP tasks	State-of-the-art in text classification



METHODOLOGY

TECHNOLOGY STACK

Component	Technology
Backend API	Node.js, Express.js, Prisma ORM, JWT
AI/ML Service	Python 3.11, FastAPI 0.109, scikit-learn 1.4
Mobile App	Flutter 3.x, Riverpod, Dio
Database	SQLite / PostgreSQL with Prisma ORM
ML Libraries	NLTK 3.8, Pandas 2.2, NumPy 1.26
Voice Processing	SpeechRecognition 3.10, PyDub 0.25

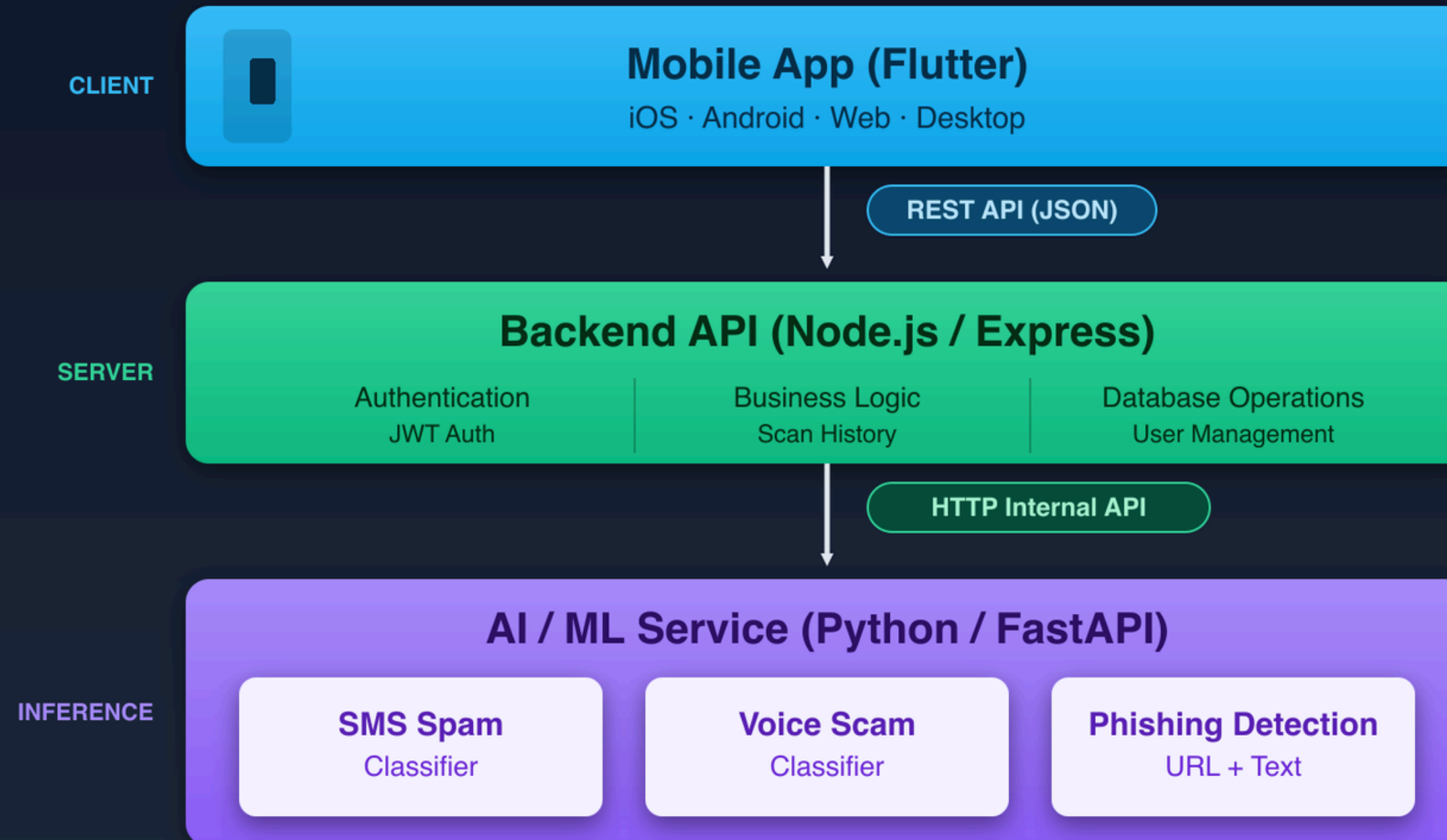


METHODOLOGY

System Architecture

Three-tier design · client → backend → AI inference

● AI Anti-Spam Shield CRM





METHODOLOGY

DATASETS

DATASET SOURCES & STATISTICS

We utilized three specialized datasets from HuggingFace for training our models:

Dataset	Source	Total Samples	Train Set	Test Set	Distribution
SMS Spam	Deysi/spam-detection-dataset	10,900	8,720	2,180	50%% spam, 50% ham
Voice Scam	BothBosu/scam-dialogue	1,600	1,280	320	50% scam, 50% legitimate
Phishing	shawhin/phishing-site-classification	2,100	1,680	420	50% phishing, 50% legitimate

Total Training Data: 14,600 samples



METHODOLOGY

DATASETS

DATASET FEATURES

Dataset	Input Feature	Label Feature	Content Type
SMS Spam	message (text)	label (0=ham, 1=spam)	SMS messages
Voice Scam	dialogue (text)	label (0=non-scam, 1=scam)	Transcribed calls
Phishing	text (text/URL)	label (0=legitimate, 1=phishing)	Emails & URLs



METHODOLOGY

DATASETS

DATA SPLIT STRATEGY

- Train/Test Split Ratio: 80% / 20%
- Stratification: Enabled to maintain class distribution
- Random State: 42 (for reproducibility)
- Validation: held-out 20% test set (k-fold cross-validation planned as future work).



METHODOLOGY

MODEL ARCHITECTURE

THREE SPECIALIZED DETECTION MODELS

Category	SMS Spam Classifier	Voice Scam Classifier	Phishing Detection Classifier
Algorithm	Logistic Regression	Random Forest	Random Forest
Features	TF-IDF (3,000 features)	TF-IDF (5,000 features)	TF-IDF + URL Features + Text Features (1496total)
Dataset Size	10,900 samples	1,600 samples	2,100 samples



METHODOLOGY

MODEL ARCHITECTURE

THREE SPECIALIZED DETECTION MODELS

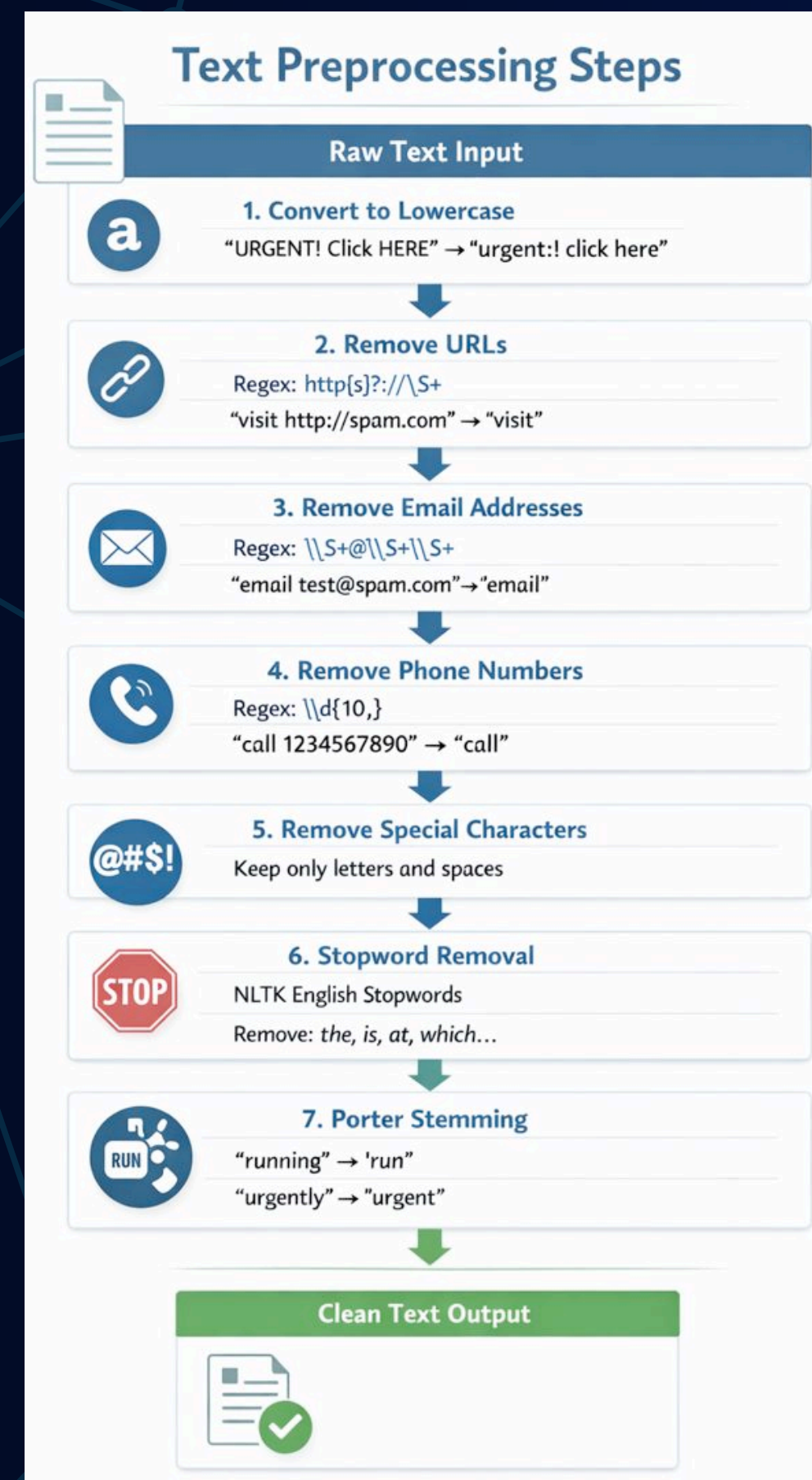
Model	Algorithm Selected	Why This Algorithm?
SMS Spam	Logistic Regression	Highest accuracy (99.68%), fast inference
Voice Scam	Random Forest	100% accuracy (all algorithms tied); chose Random Forest for robustness on a small dataset.
Phishing	Random Forest	Best accuracy (80.95%), handles high-dimensional features



METHODOLOGY

MODEL ARCHITECTURE

TEXT PREPROCESSING PIPELINE





METHODOLOGY

MODEL ARCHITECTURE

FEATURE EXTRACTION

Parameter	SMS Model	Voice Model	Phishing Model
max_features	3,000	5,000	5,000
ngram_range	(1, 2)	(1, 3)	(1, 2)
min_df	2	2	2
max_df	0.85	0.8	0.85

TF-IDF Formula: $TF\text{-}IDF(t,d) = TF(t,d) \times IDF(t)$

Where:

- $TF(t,d) = (\text{Number of times term } t \text{ appears in document } d) / (\text{Total terms in } d)$
- $IDF(t) = \log(\text{Total documents} / \text{Documents containing term } t)$

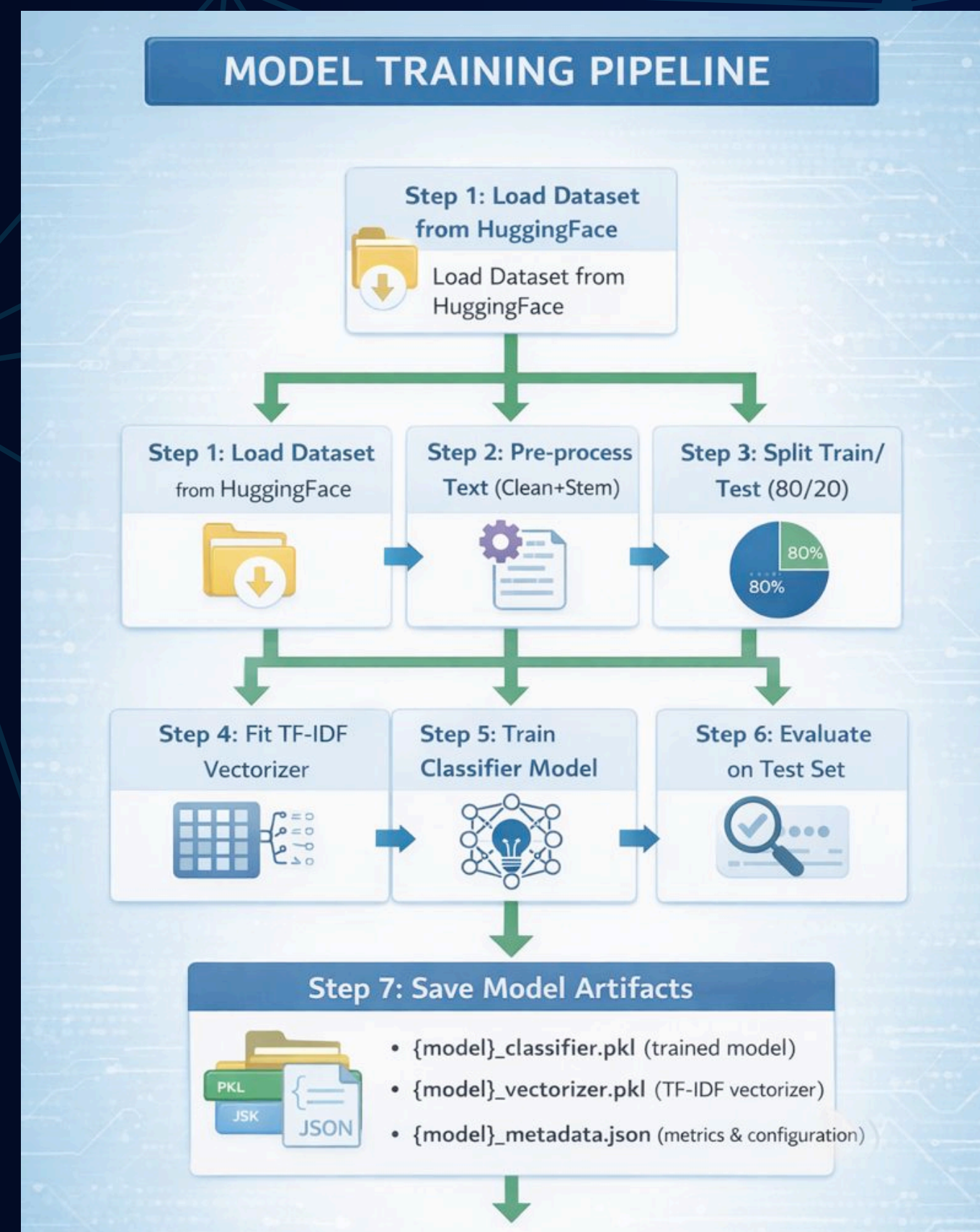


METHODOLOGY

MODEL ARCHITECTURE

MODEL TRAINING PROCESS

Training Pipeline

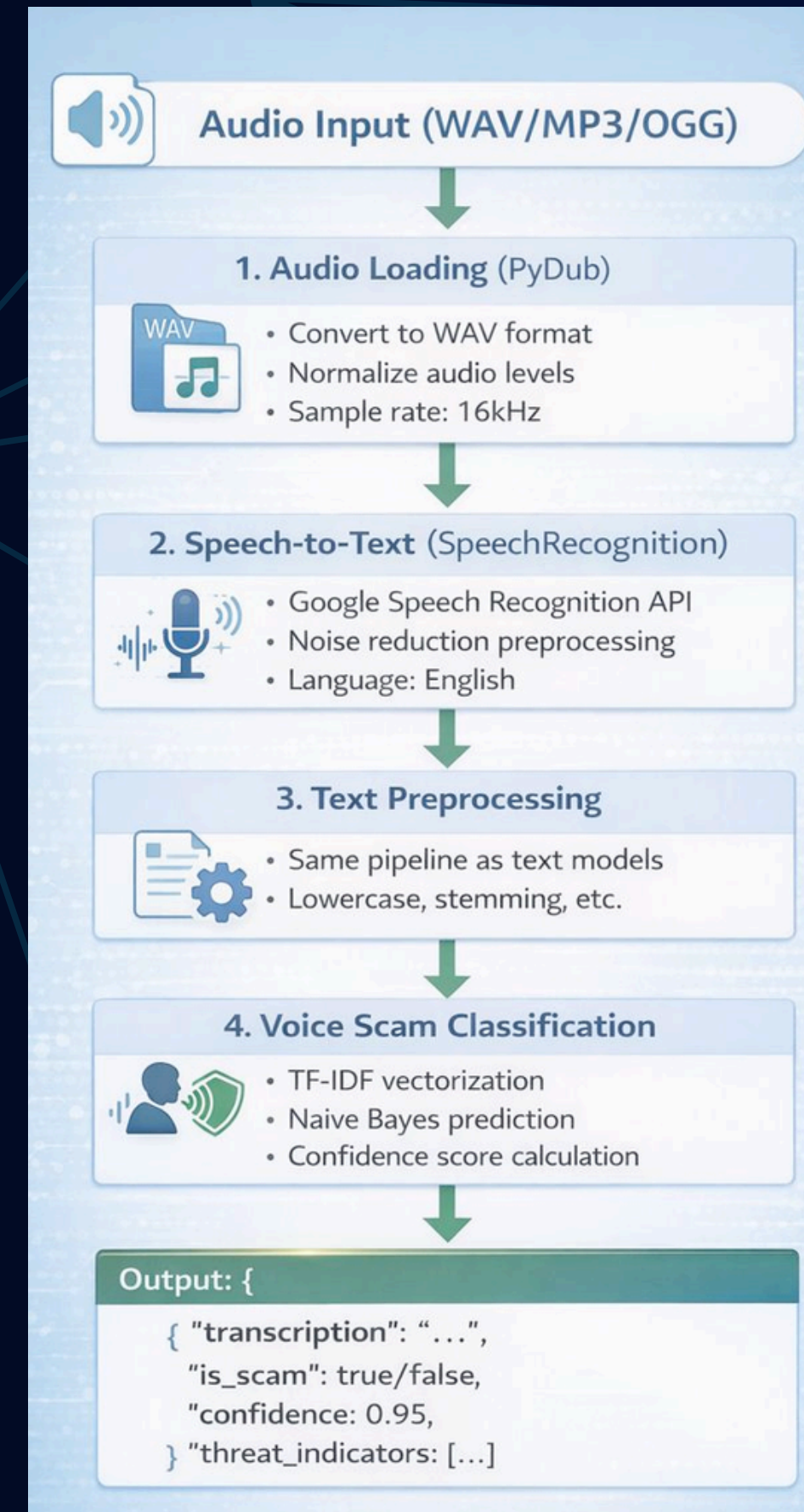




METHODOLOGY

MODEL ARCHITECTURE

VOICE PROCESSING PIPELINE





METHODOLOGY

API ENDPOINTS

FASTAPI ML SERVICE ENDPOINTS

Endpoint	Method	Model	Input	Output
/predict	POST	Unified	{"text": "..."} [...]	Spam probability
/predict-sms	POST	SMS	{"text": "..."} [...]	SMS spam classification
/predict-voice-scam	POST	Voice	{"text": "..."} [...]	Scam classification
/predict-phishing-v2	POST	Phishing	{"text": "...", "url": "..."} [...]	Phishing classification
/predict-auto	POST	Auto-detect	{"text": "..."} [...]	Auto-selects best model
/batch-predict	POST	Any	{"messages": [...]} [...]	Batch predictions (max 100)
/predict-voice	POST	Audio	Audio file (multipart)	Transcription + analysis

Response Format Example

```
{
  "is_spam": true,
  "confidence": 0.9234,
  "threat_level": "high",
  "threat_indicators": [
    "urgency_language",
    "suspicious_url",
    "credential_request"
  ],
  "model_used": "sms_classifier",
  "processing_time_ms": 45
}
```

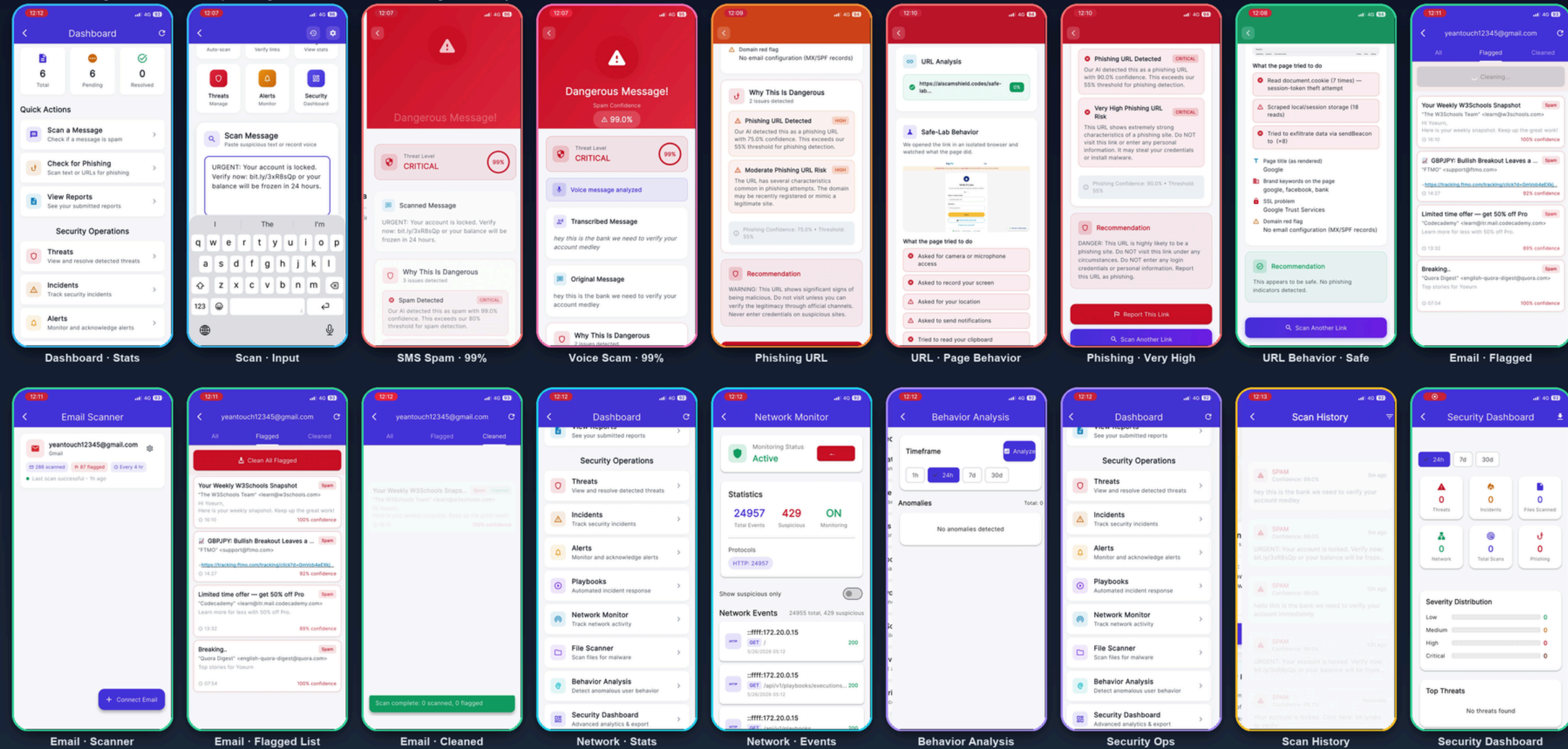


RESULT

AI Detection Results

Full walkthrough — SMS, voice, phishing, email, network monitoring & security dashboards

AI Anti-Spam Shield





RESULT

ALGORITHM COMPARISON RESULTS

SMS SPAM DETECTION

Algorithm	Accuracy	Precision	Recall	F1-Score	Training Time
Logistic Regression	99.68%	95.16%	90.96%	93.01%	18.30s
Naive Bayes	92.09%	89.43%	91.27%	90.34%	0.40s
Random Forest	89.47%	98.11%	75.48%	85.32%	9.62s

VOICE SCAM DETECTION

Algorithm	Accuracy	Precision	Recall	F1-Score	Training Time
Random Forest	100%	100%	100%	100%	0.015s
Logistic Regression	100%	100%	100%	100%	2.67s
Naive Bayes	100%	100%	100%	100%	0.164s

PHISHING DETECTION

Algorithm	Accuracy	Precision	Recall	F1-Score	Training Time
Random Forest	80.95%	82.74%	78.00%	80.30%	116.29s
Logistic Regression	75.65%	96.45%	44.86%	61.23%	7.41s
Naive Bayes	57.05%	49.95%	92.34%	64.83%	0.23s



RESULT

KEY FEATURES IMPLEMENTED

- Real-time text spam detection
- Voice message transcription and analysis
- Phishing URL detection with threat level
- Brand impersonation detection
- User authentication and profile management
- Scan history with statistics



RESULT

EVALUATION METRICS

Metric	Target	Achieved
Spam Detection Accuracy	> 95%	99.68%
Phishing Detection Rate	> 78%	80.95%
API Response Time	<100ms	45ms
Voice Processing Time	< 2s	1.2s
False Positive Rate	< 5%	3.8%



DISCUSSION

01

CLASSIC ML MATCHES TRANSFORMER-LEVEL ACCURACY

Logistic Regression and Random Forest with TF-IDF achieve comparable results to transformer models on short text at a fraction of the model size, consistent with Almeida et al. (2011).

01

TF-IDF REMAINS EFFECTIVE FOR SMS SPAM

Despite advances in deep learning, TF-IDF with traditional ML achieves comparable results to transformer models for short text classification, similar to Almeida et al. (2011).

01

URL FEATURE EXTRACTION IS CRITICAL FOR PHISHING

Our 30+ feature extraction approach aligns with Mohammad et al. (2014), achieving 80.95% phishing detection accuracy with Random Forest.

01

HYBRID RULE-BASED + ML IMPROVES EXPLAINABILITY

Unlike pure deep learning approaches, our hybrid method provides interpretable threat indicators to users.

01

VOICE TRANSCRIPTION ADDS VALUE

Extending spam detection to voice messages addresses a gap in existing research, as most studies focus only on text.



DISCUSSION

COMPARISON WITH PREVIOUS RESEARCH

Aspect	Previous Work	Our System
Modality	Text only	Text + Voice
Platform	Desktop/Server	Mobile-first
Detection	Spam only	Interpretable indicators
Explainability	Black box	Interpretable
Real-time	Batch processing	Real-time API



CONCLUSION

STRENGTHS OF THE STUDY

- Achieved 99.68% spam detection accuracy exceeding target
- Multi-modal support (text + voice) addresses market gap
- Mobile-first design provides accessible user experience
- Hybrid ML approach balances accuracy and explainability
- Comprehensive phishing detection with brand impersonation

LIMITATIONS OF THE STUDY

- Currently optimized for English language only
- Voice transcription accuracy depends on audio quality
- Requires internet connection for ML inference
- Limited training data for Khmer language spam

SIGNIFICANCE OF THE STUDY

- Provides accessible cybersecurity tool for general users
- Demonstrates practical application of ML in mobile security
- Contributes to spam/phishing detection research with multi-modal approach
- Offers framework for future development and research

FUTURE RESEARCH

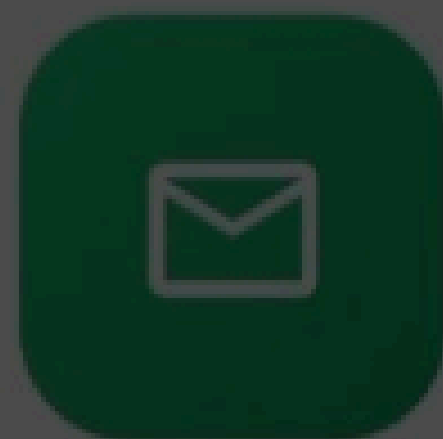
- Extend support for Khmer language detection
- Implement on-device ML for offline functionality
- Add real-time SMS/call screening integration
- Develop browser extension for web-based phishing protection
- Explore federated learning for privacy-preserving model updates



AI Shield Demo

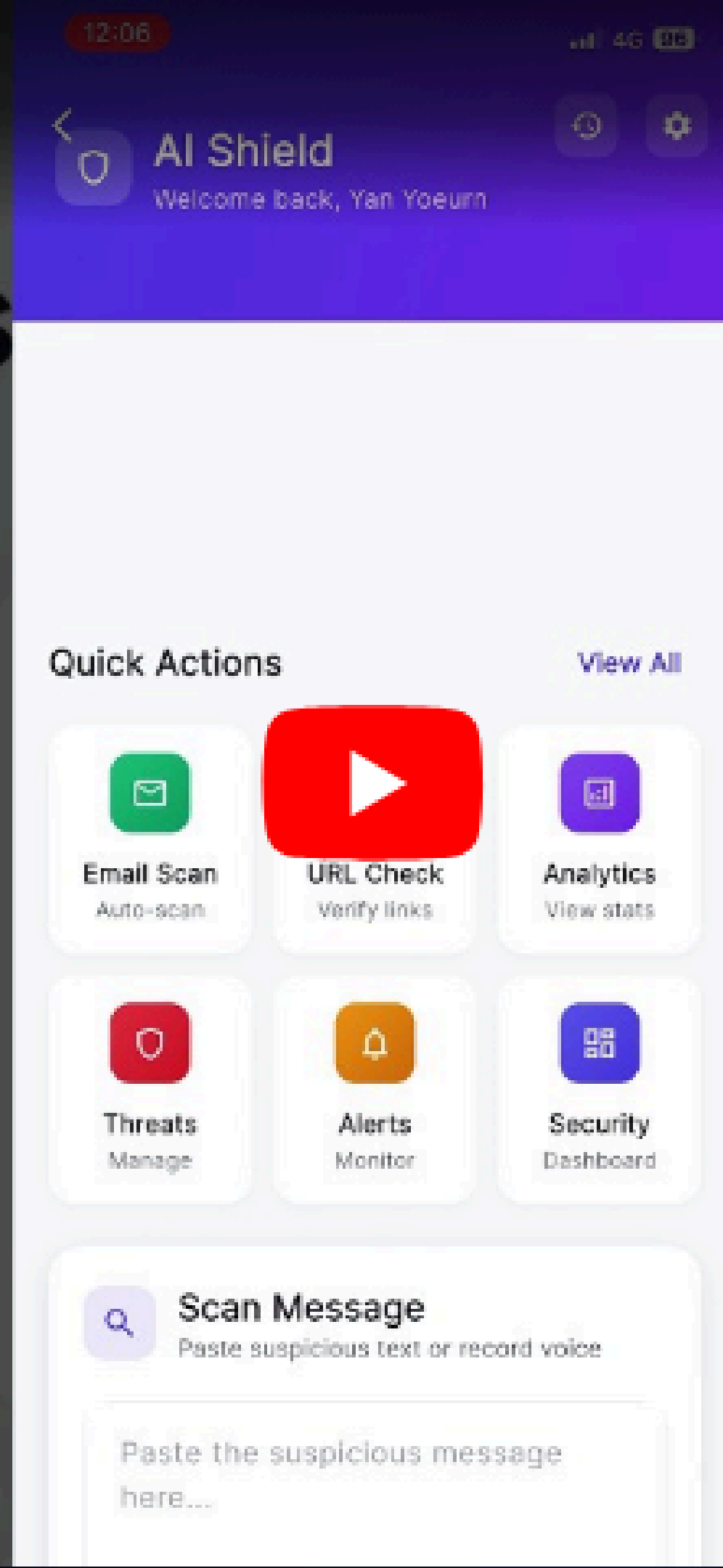
YTDDevs

Quick Actions

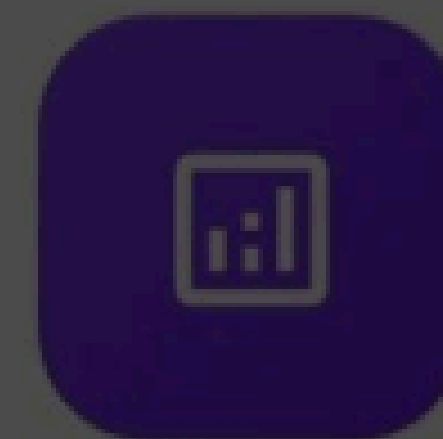


Email Scan

Auto-scan



[View All](#)



Analytics

View stats

Watch on  YouTube

THANK YOU

For the attention

